

Lovepreet Sharma

Software Engineer | Machine Learning Engineer | M.Tech Data Analytics, NIT Jalandhar

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SUMMARY

Software and ML engineer who ships tested systems: multi-tenant SaaS (PostgreSQL), a RAG platform (FastAPI) and multi-agent AI (PyTorch, IEEE paper). **GATE CS/IT qualified 2024, 2025, 2026; PPO** as an ML intern.

EDUCATION

Dr. B R Ambedkar National Institute of Technology (NIT) Jalandhar

M.Tech in Data Analytics

Jalandhar, Punjab

2025 – 2027 (*Expected*)

Giani Zail Singh Campus College of Engineering & Technology

B.Tech in Computer Science and Engineering; CGPA: 7.85/10

Bathinda, Punjab

2021 – 2025

EXPERIENCE

ThinkNEXT Technologies Pvt. Ltd.

Jan 2025 – Jun 2025

Machine Learning Engineer Intern

Mohali, Punjab

- Built end-to-end ML pipelines in Python (feature engineering, training, evaluation) using Scikit-learn and Pandas.
- Designed modular backend services and **REST APIs** serving trained models from SQL-backed data pipelines.
- Awarded a **Pre-Placement Offer (PPO)** for technical performance and project ownership.

Infowiz Pvt. Ltd.

Jun 2023 – Jul 2023

Software Development Intern

Mohali, Punjab

- Built Hospital Management System modules in PHP/MySQL with **role-based access control** and tuned SQL queries.

PROJECTS

Agentic AI for Plant Disease Diagnosis | *Python, PyTorch, Multi-Agent Systems, RAG, FAISS, LLMs* **GitHub**

- Designed a multi-agent system (vision, confidence, reasoning agents + orchestrator) over a 38-class ResNet18 ensemble that routes each case to direct diagnosis, RAG-grounded LLM reasoning, or human escalation.
- Cut the **harmful-action rate from 75.8% to 14.2%** vs. a trust-the-model baseline via ensemble disagreement.
- Added image-quality gating (blur, brightness) and grounded LLM explanations in cited sources retrieved via FAISS.

DineDirect: Multi-Tenant Restaurant SaaS | *TypeScript, Next.js, PostgreSQL, Docker* *Commercial (pre-launch)*

- Built a SaaS giving each restaurant a branded PWA storefront, owner console and admin, served by one Next.js app.
- Enforced tenant isolation with **PostgreSQL row-level security** and composite keys, verified by isolation tests.
- Built an order state machine with **optimistic locking**, idempotent placement and a transactional outbox.

RAG Engine: Document Q&A Platform | *Python, FastAPI, LangChain, FAISS, Next.js, Docker, Nginx* **GitHub**

- Built a full-stack document Q&A platform with page-level citations; API, UI and Nginx deploy via Docker Compose.
- Implemented **hybrid retrieval** (FAISS + BM25 fusion, cross-encoder reranking), evaluated with MRR and nDCG.
- Added JWT role-based auth, rate limiting, SSE streaming and a multi-provider LLM layer; **149 pytest tests**.

Loan Management System (Desktop Application) | *Node.js, Express, SQLite, CI/CD* *Commercial product*

- Shipped an offline desktop app in **daily use by a finance business** for loans, repayments and accounting.
- Implemented 4 interest models (reducing EMI, flat, fixed, zero-interest) with automatic payment allocation.
- Derived balances from an **append-only ledger** (35-assertion test suite); built installer via GitHub Actions.

PUBLICATIONS

- *Agentic AI for Plant Disease Diagnosis: An Autonomous Multi-Agent Framework Integrating Vision, Retrieval, and Reasoning.* IEEE conference paper, accepted.

TECHNICAL SKILLS

Languages: Python, C, JavaScript, TypeScript, SQL, Kotlin, PHP

AI/ML: PyTorch, Scikit-learn, Pandas, NumPy, LangChain, RAG, LLMs, Multi-Agent Systems, Federated Learning

Backend & Web: FastAPI, Node.js, Express, Next.js, React, REST APIs, JWT, SQLAlchemy

Databases & DevOps: PostgreSQL, MySQL, SQLite, FAISS, Docker, Nginx, GitHub Actions, Git, Linux, pytest

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks